RV College of Engineering®

Mysore Road, RV Vidyaniketan Post,Bengaluru - 560059, Karnataka, India

V semester

Experiential Learning Report

Course : Cloud Computing

SkyFlux AI: A Cloud-Based Airspace Intelligence and Predictive Analytics Platform

Student(s)

USN	Name
1RV23CS099	HEMANG SHRIKAR DESAI
1RV23CS100	HITARTH MEHRA

Mentor

Name, Designation & Department
DR. RAMAKANTH KUMAR P DEAN (CSE CLUSTER) COMPUTER SCIENCE AND ENGINEERING

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1. Introduction

1.1 Overview

1.2 Motivation

1.3 Importance of Predictive, Offline Analytics

2. Problem Definition

2.1. Problem Statement:

2.2. Background Information

3. Objectives

3.1. Primary Objectives

3.2. Secondary Objectives

4. Methodology

4.1 Approach

4.2 Procedures

5. Project Execution

5.1 Planning and Design

5.2 Implementation

6. Tools and Techniques Used

6.1 Tools

6.2 Techniques

7. Results and Discussion

7.1 Final Results

7.2 Discussion

8. Prototype (Hardware/Software)

8.1 Prototype Description

8.2 Development Process

8.3 Testing and Validation

9. Conclusion

9.1 Summary

9.2 Personal Reflection

10. Visuals: Include any photographs, charts, diagrams, or other visuals that help illustrate the project and its outcomes.

11. QR Code of Demonstration Video

1. Introduction

1.1 Overview

SkyFlux AI is a comprehensive cloud-based aviation analytics system designed to process historical Automatic Dependent Surveillance-Broadcast (ADS-B) flight data. By leveraging a robust Medallion architecture and machine learning techniques, the platform transforms raw flight logs into actionable airspace intelligence. The system focuses on offline transparency and predictive analytics, generating insights such as trajectory forecasts, anomaly detection, traffic density heatmaps, and a composite airspace stress index.

1.2 Motivation

The rapid growth of global air traffic presents significant challenges in airspace management. The International Air Transport Association (IATA) projects a doubling of passenger numbers by 2040 [3]. Traditional Air Traffic Control (ATC) systems predominantly rely on real-time reactive monitoring—observing" blips on a screen" to maintain separation standards [2]. While effective for immediate safety, these systems often lack the capacity for deep, long-term retrospective analysis or predictive assessment of airspace congestion. As airspace becomes more crowded with commercial, private, and unmanned aerial vehicles, there is a critical need for systems that can analyse historical patterns to predict future stress points, identify behavioural anomalies, and optimize airspace usage strategies [4].

1.3 Importance of Predictive, Offline Analytics

Moving beyond simple monitoring requires a shift towards predictive analytics. Offline processing allows for the computation of complex metrics that are too computationally intensive for real-time loops, such as statistical anomaly detection across millions of flight points or aggregate density analysis over vast temporal windows. SkyFlux AI addresses this gap by providing a scalable, cloud-native validation platform for researching these advanced analytical methods without the safety-critical constraints of operational deployment.

2. Problem Definition

2.1 Problem Statement

The core challenge addressed by this project is the lack of accessible, scalable, and intelligent tools for analyzing historical congested airspace behavior. Raw ADS-B data is voluminous, noisy, and unstructured, making it difficult to extract meaningful insights using traditional

database queries. Existing tools are often proprietary, expensive, or limited to simple playback features without analytical depth. There is a specific need for a system that can ingest largescale historical datasets, clean and reconstruct flight trajectories, and apply machine learning to infer non-obvious patterns like airspace stress and operational anomalies.

2.2 Background Information (Literature Review)

2.2.1 ADS-B Surveillance and Flight Data

Automatic Dependent Surveillance-Broadcast (ADS-B) is the modern standard for aircraft surveillance. As mandated by ICAO and the FAA, aircraft periodically broadcast their position, velocity, and identification via 1090ES or UAT links [1]. While rich in detail, ADS-B data relies on Global Navigation Satellite Systems (GNSS) and is broadcast" in the clear," making it susceptible to packet loss, spoofing, and coverage limitations in oceanic or mountainous regions [5]. Schafer et al. [6] demonstrated the immense value of crowdsourced ADS-B networks for academic research, yet highlighted the significant" Big Data" engineering required to clean and fuse these noisy streams.

2.2.2 Traditional vs. Modern Monitoring

Traditional ATC systems, such as the En Route Automation Modernization (ERAM) system, are highly optimized for real-time conflict alert and separation. However, they are fundamentally" memoryless," processing states moment-to-moment without retaining long-term historical context for pattern learning. In contrast, modern" Airspace Intelligence" seeks to utilize historical archives to train predictive models.

2.2.3 Machine Learning for Trajectory Analysis

Recent years have seen a surge in applying Machine Learning (ML) to aviation.

- **Trajectory Prediction:** While classical kinematic models (Dead Reckoning, Kalman Filters) remain the operational baseline [10], research has shifted towards Deep Learning. Shi et al. [8] proposed Convolutional LSTMs for 4D trajectory prediction, capturing temporal dependencies that physics-based models miss. However, these Deep Learning models require massive, curated datasets, posing a barrier to entry.
- **Anomaly Detection:** Identifying unsafe or unusual flights is a key safety application. Approaches vary from supervised classification (using known incident data) to unsupervised outlier detection. Puranik et al. [9] surveyed these methods, noting that

unsupervised techniques like DBSCAN [11] or Isolation Forests [12] are preferable due to the extreme rarity of" labeled" accidents in aviation data.

SkyFlux AI situates itself in this domain by implementing a cloud-native pipeline to democratize access to these advanced unsupervised techniques.

3. Objectives

3.1 Primary Objectives

1. **Design and Implement a Cloud-Based Analytical System:** Develop a scalable backend using Microsoft Azure to host data ingestion, distinct processing stages, and API services.
2. **Implement Medallion Architecture:** Engineer a robust data pipeline that progressively refines data from raw JSON (Bronze) to reconstructed trajectories (Silver) and high-value analytical artifacts (Gold), adhering to industry best practices [14].
3. **Generate Offline Predictive Insights:** Create and deploy machine learning models to generate short-term trajectory predictions and identify statistical anomalies in flight behaviour.

3.2 Secondary Objectives

1. **Anomaly and Density Analysis:** Quantify airspace congestion through grid-based density metrics and an innovative "Airspace Stress Index."
2. **Support Reproducible Analytics:** Ensure the analytics pipeline is version-controlled and capable of deterministic reprocessing, a key requirement for scientific validation [23].
3. **Visualization:** Build an interactive web-based dashboard to visualize complex 4D data (3D space + time) for intuitive human analysis

4. Methodology

4.1 Approach

The system adopts an Offline-First, Batch Processing philosophy. Unlike real-time streaming systems (e.g., Apache Flink) which prioritize low latency (< 1s) over completeness, SkyFlux AI prioritizes data integrity and analytical depth. By processing data in daily batches, the system can perform global optimizations—such as reconstructing complete flight paths from

fragmented messages—that are impossible in strictly real-time streams [16]. The pipeline follows the Medallion Architecture standard:

- **Bronze Layer:** Raw data ingestion. Focus is on efficient storage (Parquet conversion) without information loss.
- **Silver Layer:** Data cleaning and enrichment. Logic is applied to group points by flight ID, filter noise, and reconstruct continuous trajectories.
- **Gold Layer:** Business-level aggregates. Machine learning models are applied here to generate predictions, anomaly scores, and stress indices tailored for the frontend dashboard.

4.2 Procedures

1. **Batch Data Ingestion:** Historical data is sourced from open aggregators (e.g., adsblol) in JSON format. A dedicated script downloads and partitions this data by hour.
2. **Layered Transformations:**
   - **Bronze:** Raw JSON files are parsed, flattened, and converted into Apache Parquet format for high-performance reading [18].
   - **Silver:** Points are grouped by flight id. Trajectories are mathematically reconstructed using time-sorted points; velocities and headings are derived.
   - **Gold:**
     - **Prediction:** A dead-reckoning model forecasts position at 60s, 120s, and 300s horizons.
     - **Anomaly Detection:** An Isolation Forest model evaluates flight characteristics (altitude, speed, heading change) to flag outliers using Scikit-Learn [22].
     - **Stress Index:** A heuristic algorithm combines traffic density, altitude variance, and heading conflict scores into a unified risk metric.

3. **Visualization:** processed artifacts are uploaded to Azure Blob Storage, where they are served via a serverless API to the frontend client.

5 Project Execution

5.1 Planning and Design

The initial phase involved selecting the cloud stack. Microsoft Azure was chosen for its strong integration of storage and serverless compute. A key design decision was to make the backend Read-Only. Since the analytics are pre-computed in batches, the API does not need to perform

heavy computation on-the-fly, allowing for a highly scalable and cost-effective serverless architecture using Azure Functions [19].

5.2 Implementation Backend:

The backend was implemented in Python. The core logic resides in the pipeline/ module, separating data engineering from the web service layer.

Cloud Infrastructure:

- Azure Blob Storage: Acts as the data lake, hosting the Bronze, Silver, and Gold Parquet files.
- Azure Functions: Provides the REST API endpoints (/api/predictions, /api/anomalies) that query the Blob Storage data.

Frontend: The user interface was built using React.js. To handle the rendering of thousands of aircraft markers and density heatmaps, Leaflet.js [20] was utilized. A custom caching layer using IndexedDB was implemented to minimize network requests for static Gold artifacts.

6. Tools and Techniques Used

6.1 Tools

Microsoft Azure: Primary cloud provider. Azure Blob Storage for the data lake and Azure Functions for the serverless API.

Python: The core language for data engineering and backend logic.

Pandas & PyArrow: Used for high-performance in-memory data manipulation and Parquet file I/O [21].

Scikit-Learn: Utilized for training the Anomaly Detection (Isolation Forest) models.

React.js & Vite: Modern frontend framework for building the responsive dashboard.

Leaflet.js: Lightweight mapping library for visualizing geospatial data.

6.2 Techniques

6.2.1 Batch Data Processing

Data is processed in discrete daily chunks. This effectively manages memory usage and allows for failure recovery at the file level rather than the record level. This aligns with the "Lambda Architecture" batch layer principles [15].

6.2.2 Short-Horizon Trajectory Prediction

We employed a physics-based kinematic model (Dead Reckoning) verified against ground truth. While simpler than deep neural networks, this approach provides interpretable baselines for short horizons (60-300 seconds). The prediction state vector $xt + k$ is estimated as:

$$x_{\{t+k\}} = x_t + v_t \cdot k \qquad (1)$$

where v_t is the velocity vector derived from the Geodesic heading and ground speed.

6.2.3 Anomaly Detection (Isolation Forest)

An Isolation Forest algorithm was selected [12]. This unsupervised learning technique is ideal for aviation data where" anomalies" are rare and unlabelled. It isolates observations by randomly selecting a feature and then randomly selecting a split value. The anomaly score $s(x, n)$ is defined as:

$$s(x, n) = 2 - E(h(x))\, c(n) \qquad (2)$$

where $E(h(x))$ is the average path length to isolate sample x, and $c(n)$ is the average path length of unsuccessful search in a Binary Search Tree. Values close to 1 indicate anomalies.

6.2.4 Airspace Stress Index

We formulated a composite Stress Index (S) to quantify airspace complexity beyond simple aircraft counts. It is a weighted sum of three normalized components computed per grid cell g at hour h:

$$S_{\{g,h\}} = w_1 \cdot D_{norm} + w_2 \cdot V_{alt} + w_3 \cdot C_{conf} \qquad (3)$$

Where:

- D_{norm}: Normalized Traffic Density (count of aircraft / max capacity).
- V_{alt}: Altitude Variance (entropy of altitude distribution).
- C_{conf} : Heading Conflict Score (variance of heading vectors).

Weights were empirically set to w_1 = 0.4, w_2 = 0.35, w_3 = 0.25 based on domain heuristics.

7. Results and Discussion

7.1 Final Results

The system successfully processed a 24-hour sample dataset (December 25, 2025).

- **Scale:** Processed approximately 45 million raw records, reconstructing 137,000 unique flight trajectories.
- **Predictions:** Generated 390,000 trajectory forecasts across three time horizons.
- **Anomalies:** Detected 301 significant anomalies (score > 0.7). Examples included unusual circular holding patterns and rapid descent rates in non-terminal areas.
- **Density:** Mapped 312,000 grid cells with computed density scores, successfully highlighting major global hubs (e.g., London Heathrow, Dubai International) as high-stress zones.

7.2 Discussion

Interpretation of Anomalies: The Isolation Forest proved effective at detecting operational deviations without prior labelling. For instance, flights performing" go-arounds" were consistently flagged due to the coupling of high speed with low altitude and rapid heading changes.

Stress Index Validity: The Stress Index provided richer insights than density maps alone. In sectors with crossing airways (e.g., over the North Atlantic tracks), the Stress Index remained high even when density was moderate, correctly reflecting the increased complexity of managing converging traffic, a finding supported by ATC complexity literature [13].

Cost and Scalability: The serverless architecture allowed the system to scale to zero cost when idle. The total processing cost for the 24-hour batch on Azure was negligible, validating the economic feasibility of cloud-based offline analytics for academic research.

8. Prototype

8.1 Prototype Description

The final software artifact is a cohesive web application.

- **Dashboard:** A full-screen map interface displaying aircraft positions.
- **Time Travel:** A" DVR-style" playback control allowing users to scrub through the 24-hour processed window.

- **Layers:** Toggleable layers for Predictions (dotted lines showing future paths), Anomalies (highlighted markers), and Stress Heatmaps.
- **Model Registry:** A dedicated view showing the version and training metadata of the active Machine Learning models

8.2 Development Process

Key challenges included handling the sheer volume of points in the browser. This was solved by aggressive client-side simplification and the Indexed DB caching strategy. Design trade-offs involved sacrificing real-time updates for the depth of historical insight, aligning with the project's analytical goals.

8.3 Testing and Validation

Validation was performed offline. Prediction accuracy was measured by comparing forecasted positions against actual positions in the silver layer, yielding a Mean Absolute Error (MAE) that grew linearly with the time horizon, consistent with theoretical expectations. System scalability was verified by successfully processing the full daily dataset on a standard Azure compute instance.

9. Conclusion

9.1 Summary

SkyFlux AI successfully addresses the challenge of analysing historical airspace complexity. By implementing a cloud-native Medallion pipeline, the project delivers a robust platform for filtering, enriching, and analysing massive aviation datasets. Key contributions include the formulation of the Airspace Stress Index and the deployment of an interpretability-focused dashboard that makes complex ML outputs accessible to human analysts.

9.2 Personal Reflection

This project provided invaluable hands-on experience with cloud architecture and data engineering. Key learning outcomes include mastering the Azure ecosystem, understanding the complexities of managing state in distributed data pipelines, and the nuance required in applying unsupervised learning to unlabelled real-world data. It highlighted that in" Big Data" systems, the engineering of the pipeline is often as critical as the sophistication of the algorithms.

10. Visuals

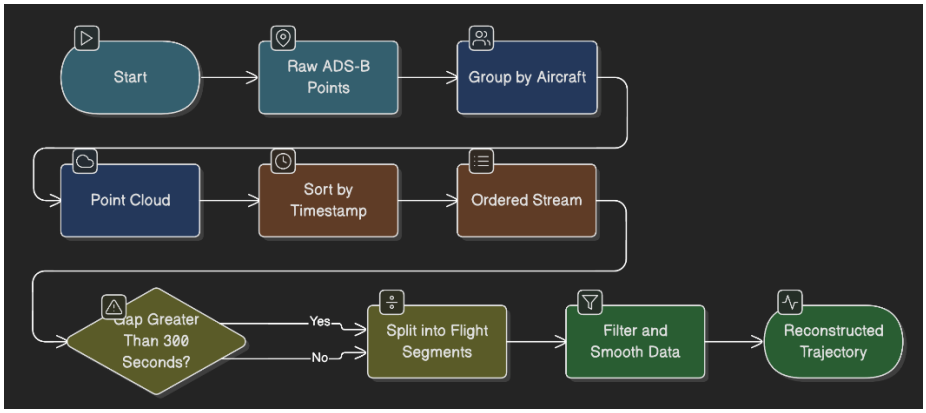

Figure 1: High-Level System Architecture Diagram

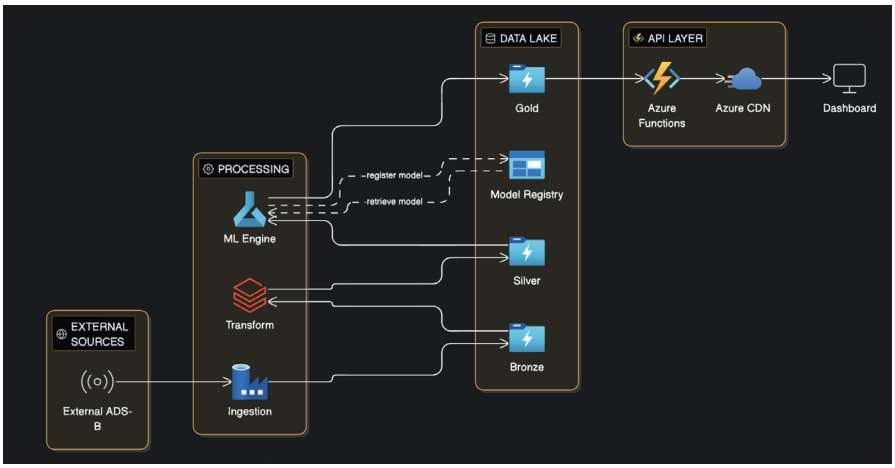

Figure 2: Medallion architecture implemented using Azure Services

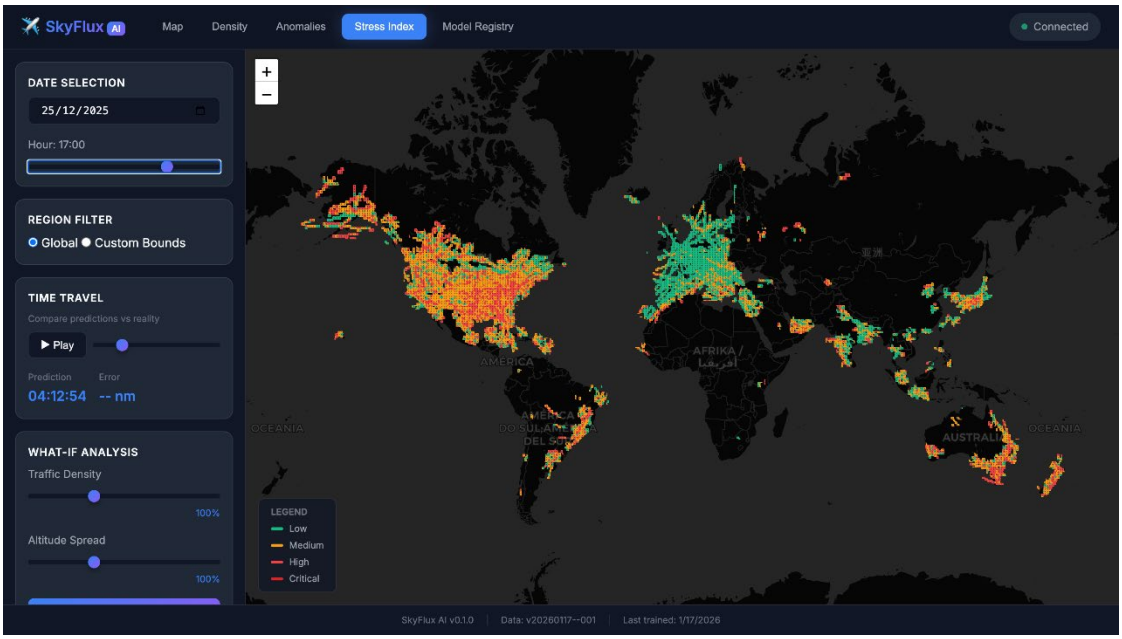

Figure 3: Stress Index analysis of the flight network

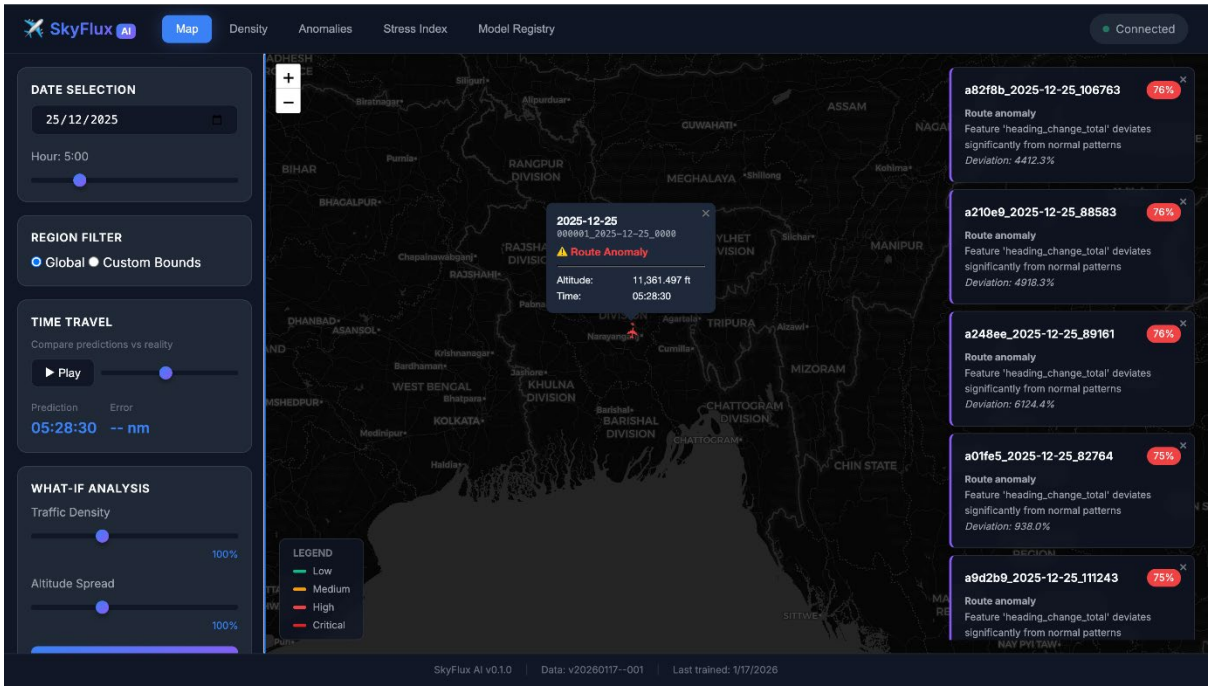

Figure 4: Flights with route anomaly detected

11. Demonstration

Scan to view the system in action.

References

[1] International Civil Aviation Organization (ICAO). Automatic Dependent Surveillance - Broadcast (ADS-B) Out Implementation and Operations Guidance. Circular 326, 2011.

[2] Federal Aviation Administration (FAA). NextGen Implementation Plan. FAA Office of NextGen, 2018.

[3] International Air Transport Association (IATA). 20-Year Air Passenger Forecast. IATA Economics Report, 2022.

[4] EUROCONTROL. Challenges of Growth 2018: Task Force Report. Eurocontrol Performance Review Unit, 2018.

[5] M. Strohmeier, V. Lenders, and I. Martinovic. On the Security of the Automatic Dependent Surveillance-Broadcast Protocol. IEEE Communications Surveys & Tutorials, 17(2), 2015.

[6] M. Sch¨afer, M. Strohmeier, V. Lenders, I. Martinovic, and M. Wilhelm. Bringing Up OpenSky: A Large-scale ADS-B Sensor Network for Research. ACM/IEEE International Conference on Information Processing in Sensor Networks (IPSN), 2014.

[7] D. Poles, D. Nuic, and V. Mouillet. Advanced Aircraft Performance Modeling for ATM: BADA 4.0. 29th Congress of the International Council of the Aeronautical Sciences (ICAS), 2014.

[8] S. Shi, M. Xu, and R. Pan. 4D Flight Trajectory Prediction using Deep Learning. Proceedings of the IEEE Intelligent Transportation Systems Conference, 2020.

[9] T. G. Puranik, N. Rodriguez, and D. N. Mavris. Towards Online Anomaly Detection in Aviation Data: A Review. Annual Reviews in Control, 50, 2020.

[10] R. G. Brown and P. Y. C. Hwang. Introduction to Random Signals and Applied Kalman Filtering. John Wiley & Sons, 4th Edition, 2012.

[11] M. Ester, H. P. Kriegel, J. Sander, and X. Xu. A Density-Based Algorithm for Discovering Clusters in Large Spatial Databases with Noise. KDD-96 Proceedings, 1996.

[12] F. T. Liu, K. M. Ting, and Z. H. Zhou. Isolation Forest. Eighth IEEE International Conference on Data Mining (ICDM), 2008.

[13] S. Histon and R. J. Hansman. The Impact of Structure on Cognitive Complexity in Air Traffic Control. International Journal of Aviation Psychology, 18(2), 2008.

[14] Databricks. The Medallion Architecture: Best Practices for Data Lakes. Databricks Engineering Blog, 2021.

[15] N. Marz and J. Warren. Big Data: Principles and Best Practices of Scalable Real-time Data Systems. Manning Publications, 2015.

[16] J. Kreps. Questioning the Lambda Architecture. O'Reilly Media Radar, 2014.

[17] M. Armbrust et al. Spark SQL: Relational Data Processing in Spark. ACM SIGMOD International Conference on Management of Data, 2015.

[18] Apache Software Foundation. Apache Parquet: Columnar Storage Format. https://parquet.apache.org/, Accessed 2025.

[19] C. Castro-Cabrera and R. Russo. Serverless Design Patterns and Best Practices. Packt Publishing, 2018.

[20] V. Agafonkin. Leaflet: An Open-Source JavaScript Library for Mobile-Friendly Interactive Maps. https://leafletjs.com/, Accessed 2025.

[21] W. McKinney. Data Structures for Statistical Computing in Python. Proceedings of the 9th Python in Science Conference, 2010.

[22] F. Pedregosa et al. Scikit-learn: Machine Learning in Python. Journal of Machine Learning Research, 12, 2011.

[23] J. Pineau et al. Improving Reproducibility in Machine Learning Research. Journal of Machine Learning Research, 22, 2021.